

Executive Summary

PASS RATE 30% target 90%	TOTAL TESTS 168 executed	PASSED 51 30% of total	FAILED 117 70% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing reveals significant reliability concerns with a 30% overall pass rate against the 90% target, indicating widespread endpoint instability and response inconsistencies across both API automation (27.5% pass rate) and smoke validation (33% pass rate) suites. The system exhibits substantial functional gaps with 113 total failures suggesting systemic issues in API contract adherence, data validation, error handling, or service availability that require immediate backend investigation and remediation to achieve production readiness.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	29	59	33%	FAIL
TOTAL		168	51	117	30%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#66	15 Jul	168	30%	117	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	51	30.4%	180ms	Within SLA
Requests Failed	117	69.6%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

QUALITY INDICATORS

Severity	Count	%	Indicator	Value	Target
Critical	6	5%	Endpoint Coverage	88%	85%
High	8	7%	Execution Rate	100%	100%
Medium	4	3%	Contract Validation	95%	90%
Low	2	2%	Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate backend service health assessment focusing on the 113 failing endpoints to identify common failure patterns	Immediate	P1
2	Prioritize fixing critical revenue calculation and data persistence APIs that impact core business functionality	3 days	P1
3	Review API response schemas and error handling mechanisms to ensure compliance with documented specifications	5 days	P2
4	Implement comprehensive backend logging and monitoring to capture failure root causes during test execution	1 week	P2
5	Establish backend service dependency mapping to identify potential cascading failures or integration bottlenecks	1 week	P2
6	Execute performance profiling on high-failure endpoints to detect timeout, throttling, or resource constraint issues	2 weeks	P3

KEY FINDINGS

1	Critical pass rate deficit of 60 percentage points below target threshold indicates extensive backend service issues
2	API automation suite showing 58 endpoint failures suggests widespread request/response contract violations or service unavailability
3	Backend smoke validation failures (59 tests) indicate core system functionality and integration points are unstable
4	Consistent failure pattern across both test suites points to fundamental backend architectural or deployment issues
5	Low pass rates across all test categories suggest systemic problems rather than isolated endpoint defects

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

